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Chapter 7

Percent and Percentage

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Percent and Percentage

Exercise 7(A)

Question 1(i)

A bag contains 8 white and 12 black balls, the percentage of black balls in the bag is:

1. 20%

2. 150%

3. 60%



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4. 40%

Answer

Given:

Number of white balls = 8

Number of black balls = 12

Total number of balls = $12 + 8 = 20$

$$\text{Percentages of black balls} = \frac{\text{Number of black balls}}{\text{Total number of balls}} \times 100 \%$$

$$= \frac{12}{20} \times 100 \%$$

$$= \frac{1200}{20} \%$$

$$= 60 \%$$

Hence, option 3 is the correct option.

Question 1(ii)

A number is first increased by 20%, then the resulting number is decreased by 20%. On the whole the original number is increased/decreased by :

1. 4% increased
2. 4% decreased
3. 2% increased
4. 2% decreased

Answer

Let the original number be 100.

After an increase of 20 %, it becomes = $100 + 20 \% \text{ of } 100$

$$= 100 + \frac{20}{100} \times 100$$

$$= 100 + \frac{2000}{100}$$

$$= 100 + 20$$

$$= 120$$

Now it is decreased by 20%, it becomes = $120 - 20 \% \text{ of } 120$

$$= 120 - \frac{20}{100} \times 120$$

$$= 120 - \frac{2,400}{100}$$

$$= 120 - 24$$

$$= 96$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 96 - 100$$

$$= -4$$

Change on the whole is negative means change on the whole the number is decreasing.

$$\therefore \text{Percentage change} = \frac{\text{Change on the whole}}{\text{Initial value}} \times 100 \%$$

$$= \frac{4}{100} \times 100 \%$$

$$= 4\%$$

Hence, option 2 is the correct option.

Question 1(iii)

After paying 20% of the income, a man is left with ₹ 160; then the income of the man is:

1. ₹ 128
2. ₹ 192
3. ₹ 200
4. none of these

Answer

Let x be the income of a man.

The man spent 20 % of the income.

$\Rightarrow$ He saves $(100 - 20)\% = 80 \%$ of his income

So, 80 % of his income = ₹ 160

$$\Rightarrow \frac{80}{100} \times x = 160$$

$$\Rightarrow \frac{80x}{100} = 160$$

$$\Rightarrow \frac{4x}{5} = 160$$

$$\Rightarrow x = \frac{160 \times 5}{4}$$

$$\Rightarrow x = \frac{800}{4}$$

$$\Rightarrow x = 200$$

Hence, option 3 is the correct option.

Question 1(iv)

If $A + B + C = 400$ in which A is 40% and B is 45%, then the exact quantity of C in the whole is :

1. 160

2. 60

3. 180

4. 15%

Answer

The whole value is 400.

$A = 40\% \text{ of } 400$

$$A = \frac{40}{100} \times 400$$

$$A = \frac{16,000}{100}$$

$$A = 160$$

$$B = 45 \% \text{ of } 400$$

$$B = \frac{45}{100} \times 400$$

$$B = \frac{18,000}{100}$$

$$B = 180$$

$$\text{As we know, } A + B + C = 400$$

$$\Rightarrow 160 + 180 + C = 400$$

$$\Rightarrow 340 + C = 400$$

$$\Rightarrow C = 400 - 340$$

$$\Rightarrow C = 60$$

Hence, option 2 is the correct option.

Question 1(v)

A number is decreased by 20%. If the resulting number is 800, the original number is :

1. 640

2. 960

3. 1000

4. 600

Answer

Let the number be x .

After a decrease of 20%, it becomes = 800

$$\Rightarrow x - \frac{20}{100} \times x = 800$$

$$\Rightarrow x - \frac{20x}{100} = 800$$

$$\Rightarrow x - 0.2x = 800$$

$$\Rightarrow 0.8x = 800$$

$$\Rightarrow x = \frac{800}{0.8}$$

$$\Rightarrow x = \frac{8,000}{8}$$

$$\Rightarrow x = 1,000$$

Hence, option 3 is the correct option.

Question 2(i)

Evaluate :

55% of 160 + 24% of 50 - 36% of 150

Answer

55% of 160 + 24% of 50 - 36% of 150

$$\Rightarrow \frac{55}{100} \text{ of } 160 + \frac{24}{100} \text{ of } 50 - \frac{36}{100} \text{ of } 150$$

$$\Rightarrow \frac{8,800}{100} + \frac{1,200}{100} - \frac{5,400}{100}$$

$$\Rightarrow 88 + 12 - 54$$

$$\Rightarrow 100 - 54$$

$$\Rightarrow 46$$

$$\mathbf{55\% \text{ of } 160 + 24\% \text{ of } 50 - 36\% \text{ of } 150 = 46}$$

Question 2(ii)

Evaluate :

$$9.3\% \text{ of } 500 - 4.8\% \text{ of } 250 - 2.5\% \text{ of } 240$$

Answer

$$9.3\% \text{ of } 500 - 4.8\% \text{ of } 250 - 2.5\% \text{ of } 240$$

$$\Rightarrow \frac{9.3}{100} \text{ of } 500 - \frac{4.8}{100} \text{ of } 250 - \frac{2.5}{100} \text{ of } 240$$

$$\Rightarrow \frac{4,650}{100} - \frac{1,200}{100} - \frac{600}{100}$$

$$\Rightarrow 46.5 - 12 - 6$$

$$\Rightarrow 34.5 - 6$$

$$\Rightarrow 28.5$$

$$\mathbf{9.3\% \text{ of } 500 - 4.8\% \text{ of } 250 - 2.5\% \text{ of } 240 = 28.5}$$

Question 3(i)

A number is increased from 125 to 150; find the percentage increase.

Answer

Given:

Initial Number = 125

Final Number = 150

Increase in number = Final number - Initial number

$$= 150 - 125 = 25$$

$$\text{Percentage increase} = \frac{\text{Increase in number}}{\text{Initial Number}} \times 100 \%$$

$$= \frac{25}{125} \times 100 \%$$

$$= \frac{2,500}{125} \%$$

$$= 20\%$$

The percentage increase = 20 %.

Question 3(ii)

A number is decreased from 125 to 100; find the percentage decrease.

Answer

Given:

Initial Number = 125

Final Number = 100

Decrease in number = Initial number - Final number

Decrease in number = $125 - 100 = 25$

Percentage decrease = $\frac{\text{Decrease in number}}{\text{Initial Number}} \times 100 \%$

$$= \frac{25}{125} \times 100 \%$$

$$= \frac{2,500}{125} \%$$

$$= 20\%$$

The percentage decrease = 20%.

Question 4(i)

Find :

45 is what percent of 54 ?

Answer

Let 45 be x of 54

Hence,

$$x \text{ of } 54 = 45$$

$$\Rightarrow \frac{x}{100} \times 54 = 45$$

$$\Rightarrow \frac{54x}{100} = 45$$

$$\Rightarrow x = \frac{45 \times 100}{54}$$

$$\Rightarrow x = \frac{4,500}{54}$$

$$\Rightarrow x = \frac{250}{3}$$

$$\Rightarrow x = 83\frac{1}{3}$$

45 is $83\frac{1}{3}$ of 54.

Question 4(ii)

Find :

2.7 is what percent of 18 ?

Answer

Let 2.7 be x of 18

Hence,

$$x\% \text{ of } 18 = 2.7$$

$$\Rightarrow \frac{x}{100} \times 18 = 2.7$$

$$\Rightarrow \frac{18x}{100} = \frac{27}{10}$$

$$\Rightarrow x = \frac{27 \times 100}{10 \times 18}$$

$$\Rightarrow x = \frac{2,700}{180}$$

$$\Rightarrow x = 15 \%$$

2.7 is 15% of 18.

Question 5(i)

252 is 35% of a certain number, find the number.

Answer

Let the number be x . Hence,

$$35\% \text{ of } x = 252$$

$$\Rightarrow \frac{35}{100} \times x = 252$$

$$\Rightarrow \frac{35x}{100} = 252$$

$$\Rightarrow \frac{7x}{20} = 252$$

$$\Rightarrow x = \frac{252 \times 20}{7}$$

$$\Rightarrow x = \frac{5,040}{7}$$

$$\Rightarrow x = 720$$

The number will be 720.

Question 5(ii)

If 14% of a number is 315; find the number.

Answer

Let the number be x . Hence,

$$14\% \text{ of } x = 315$$

$$\Rightarrow \frac{14}{100} \times x = 315$$

$$\Rightarrow \frac{14x}{100} = 315$$

$$\Rightarrow \frac{7x}{50} = 315$$

$$\Rightarrow x = \frac{315 \times 50}{7}$$

$$\Rightarrow x = \frac{15,750}{7}$$

$$\Rightarrow x = 2,250$$

The number will be 2,250.

Question 6

Find the percentage change, when a number is changed from :

(i) 80 to 100

(ii) 100 to 80

(iii) 6.25 to 7.50

Answer

(i) 80 to 100

Given:

Initial Number = 80

Final Number = 100

Increase in number = Final number - Initial number

Increase in number = $100 - 80 = 20$

$$\text{Percentage Increase} = \frac{\text{Increase in number}}{\text{Initial Number}} \times 100 \%$$

$$= \frac{20}{80} \times 100 \%$$

$$= \frac{2,000}{80} \%$$

$$= 25\%$$

The percentage increase = 25 %.

(ii) 100 to 80

Given:

Initial Number = 100

Final Number = 80

Decrease in number = Initial number - Final number

Decrease in number = $100 - 80 = 20$

$$\text{Percentage Decrease} = \frac{\text{Decrease in number}}{\text{Initial Number}} \times 100 \%$$

$$= \frac{20}{100} \times 100 \%$$

$$= \frac{2,000}{100} \%$$

$$= 20\%$$

The percentage decrease = 20 %.

(iii) 6.25 to 7.50

Given:

Initial Number = 6.25

Final Number = 7.50

Increase in number = Final number - Initial number

Increase in number = 7.50 - 6.25 = 1.25

$$\text{Percentage Increase} = \frac{\text{Increase in number}}{\text{Initial Number}} \times 100 \%$$

$$= \frac{1.25}{6.25} \times 100 \%$$

$$= \frac{125}{625} \%$$

$$= \frac{12,500}{625} \%$$

$$= 20\%$$

The percentage increase = 20 %.

Question 7

An auctioneer charges 8% for selling a house. If the house is sold for ₹ 2,30,500. Find the charges of the auctioneer.

Answer

Given:

Selling price of the house = ₹ 2,30,500

Auctioneer charges = 8 % of the selling price

Auctioneer charges = 8 % of ₹ 2,30,500

$$= \frac{8}{100} \times 2,30,500$$

$$= \frac{18,44,000}{100}$$

$$= 18,440$$

The charges of the auctioneer = ₹ 18,440

Question 8

Out of 800 oranges, 50 are found rotten. Find the percentage of good oranges.

Answer

Given:

Total number of oranges = 800

Number of rotten oranges = 50

Number of good oranges = 800 - 50 = 750

$$\text{Percentages of good oranges} = \frac{\text{Number of good oranges}}{\text{Total number of oranges}} \times 100 \%$$

$$= \frac{750}{800} \times 100 \%$$

$$= \frac{75,000}{800} \%$$

$$= \frac{375}{4} \%$$

$$= 93\frac{3}{4} \%$$

$$\text{The percentage of good oranges} = 93\frac{3}{4} \%$$

Question 9

A cistern contains 5 thousand litres of water. If 6% water is leaked, find how many litres of water would be left in the cistern.

Answer

Given:

A cistern contains water = 5,000 l

Percentage of water leaked = 6%

Percentage of water retained = (100 % - 6 %) = 94 %

Total water retained = 94 % of 5,000 l

$$= \frac{94}{100} \times 5,000$$

$$= \frac{4,70,000}{100}$$

$$= 4,700 \text{ l}$$

Water left in the cistern would be = 4,700 l

Question 10

A man spends 87% of his salary. If he saves ₹ 325; find his salary.

Answer

Given:

A man spends salary = 87 %

The man saves = ₹ 325

The man saves = (100 % - 87 %) = 13 %

Let of the man be x .

Hence,

13 % of the salary = ₹ 325

$$\Rightarrow 13 \% \text{ of } x = ₹ 325$$

$$\Rightarrow \frac{13}{100} \times x = 325$$

$$\Rightarrow \frac{13x}{100} = 325$$

$$\Rightarrow x = \frac{325 \times 100}{13}$$

$$\Rightarrow x = \frac{32,500}{13}$$

$$\Rightarrow x = ₹ 2,500$$

The salary of man = ₹ 2,500

Question 11(i)

A number 3.625 is wrongly read as 3.265; find the percentage error.

Answer

Given:

Incorrect number = 3.265

Correct number = 3.625

Difference = Correct number - Incorrect number

Difference = 3.625 - 3.265

Difference = 0.36

$$\text{Percentage error} = \frac{\text{difference}}{\text{correct number}} \times 100 \%$$

$$= \frac{0.36}{3.625} \times 100 \%$$

$$= \frac{36}{3.625} \%$$

$$= \frac{36,000}{3,625} \%$$

$$= 9.93 \%$$

The percentage error = 9.93%.

Question 11(ii)

A number 5.78×10^3 is wrongly written as 5.87×10^3 ; find the percentage error.

Answer

Given:

$$\text{Incorrect number} = 5.87 \times 10^3 = 5,870$$

$$\text{Correct number} = 5.78 \times 10^3 = 5,780$$

$$\text{Difference} = \text{Correct number} - \text{Incorrect number}$$

$$\text{Difference} = 5,780 - 5,870$$

$$\text{Difference} = -90$$

(Negative sign shows that value is decreasing.)

$$\text{Percentage error} = \frac{\text{difference}}{\text{correct number}} \times 100 \%$$

$$= \frac{90}{5,780} \times 100 \%$$

$$= \frac{9,000}{5,780} \%$$

$$= \frac{900}{578} \%$$

$$= 1.56 \%$$

The percentage error = 1.56%.

Question 12

In an election between two candidates, one candidate secured 58% of the votes polled and won the election by 18,336 votes. Find the total number of votes polled and the votes secured by each candidate.

Answer

Given:

Winner won by = 18,336 votes

Votes secured by one candidate = 58 %

Votes secured by other candidate = (100 % - 58 %) = 42 %

Let the total number of votes be x .

Difference in votes of two candidates = 18,336

$$\Rightarrow 58\% \text{ of } x - 42\% \text{ of } x = 18,336$$

$$\Rightarrow (58\% - 42\%) \text{ of } x = 18,336$$

$$\Rightarrow 16\% \text{ of } x = 18,336$$

$$\Rightarrow \frac{16}{100} \times x = 18,336$$

$$\Rightarrow x = \frac{18,336 \times 100}{16}$$

$$\Rightarrow x = \frac{18,33,600}{16}$$

$$\Rightarrow x = 1, 14, 600$$

Votes secured by one candidate = 58 % of total votes

$$= \frac{58}{100} \times 1, 14, 600$$

$$= \frac{66, 46, 800}{100}$$

$$= 66, 468$$

Votes secured by other candidate = 42 % of total votes

$$= \frac{42}{100} \times 1, 14, 600$$

$$= \frac{48, 13, 200}{100}$$

$$= 48, 132$$

The total number of votes = 1,14,600, votes secured by one candidates = 66,468 and votes secured by other candidates = 48,132.

Question 13

In an election between two candidates, one candidate secured 47% of votes polled and lost the election by 12,366 votes. Find the total votes polled and the votes secured by the winning candidate.

Answer

Given:

Winner lost by = 12,336 votes

Votes secured by one candidate = 47 %

Votes secured by other candidate = (100 % - 47 %) = 53 %

Let the total number of votes be x .

Difference in votes of two candidates = 12,336

$$\Rightarrow 53\% \text{ of } x - 47\% \text{ of } x = 12,336$$

$$\Rightarrow (53\% - 47\%) \text{ of } x = 12,336$$

$$\Rightarrow 6\% \text{ of } x = 12,336$$

$$\Rightarrow \frac{6}{100} \times x = 12,336$$

$$\Rightarrow x = \frac{12,336 \times 100}{6}$$

$$\Rightarrow x = \frac{12,33,600}{6}$$

$$\Rightarrow x = 2,06,100$$

Votes secured by candidates who won the election = 53 % of total votes

$$= \frac{53}{100} \times 2,06,100$$

$$= \frac{1,09,23,300}{100}$$

$$= 1,09,233$$

The total number of votes = 2,06,100 and votes secured by the winning candidates = 1,09,233.

Question 14

The cost of a scooter depreciates every year by 15% of its value at the beginning of the year. If the present cost of the scooter is ₹ 8,000, find its cost :

(i) after one year

(ii) after 2 years.

Answer**Given:**

Cost of the scooter = ₹ 8,000

Depreciation in cost of scooter in 1st year = 15%

(i) After one year

Depreciation in cost of scooter = 15 % of 8,000

$$= \frac{15}{100} \times 8,000$$

$$= \frac{1,20,000}{100}$$

$$= 1,200$$

Cost of the scooter after depreciation = Original Cost - Depreciation

$$= ₹ (8,000 - 1,200)$$

$$= ₹ 6,800$$

Cost of scooter after one year = ₹ 6,800.

(ii) After 2 years.

Depreciation in cost of scooter after 2 years = 15 % of 6,800

$$= \frac{15}{100} \times 6,800$$

$$= \frac{1,02,000}{100}$$

$$= 1,020$$

Cost of the scooter after depreciation = Cost after 1 year - Depreciation

$$= ₹ (6,800 - 1,020)$$

$$= ₹ 5,780$$

Cost of scooter after two year = ₹ 5,780.

Question 15

In an examination, the pass mark is 40%. If a candidate gets 65 marks and fails by 3 marks; find the maximum marks.

Answer

Given:

Passing marks = 40%

Marks scored by candidate = 65 marks

Marks candidate fails by = 3 marks

Let x be the maximum marks.

$$\begin{aligned}\therefore \text{Passing marks} &= \frac{40}{100} \times x \\ &= \frac{2}{5} \times x \\ &= \frac{2x}{5} \dots\dots\dots(1)\end{aligned}$$

Since, the candidate got 65 marks and fails by 3 marks. Hence, we can say that
passing marks = $(65 + 3) = 68$ marks.

Using equation (1), we get,

$$\begin{aligned}\Rightarrow \frac{2}{5} \times x &= 68 \\ \Rightarrow x &= \frac{68 \times 5}{2} \\ \Rightarrow x &= \frac{340}{2} \\ \Rightarrow x &= 170\end{aligned}$$

The maximum marks = 170.

Question 16

In an examination, a candidate secured 125 marks and failed by 15 marks. If the pass percentage was 35%, find the maximum marks.

Answer

Given:

Passing marks = 35%

A candidate gets = 125 marks

The candidate fails by = 15 marks

Let x be the maximum marks.

$$\therefore \text{Passing marks} = \frac{35}{100} \times x$$

$$= \frac{7}{20} \times x$$

$$= \frac{7x}{20} \dots\dots\dots(1)$$

Since the candidate got 65 marks and fails by 15 marks. Hence, we can say that
passing marks = $(125 + 15) = 140$ marks.

Using equation (1), we can say

$$\Rightarrow \frac{7x}{20} = 140$$

$$\Rightarrow x = \frac{140 \times 20}{7}$$

$$\Rightarrow x = \frac{2800}{7}$$

$$\Rightarrow x = 400$$

The maximum marks = 400.

Question 17

In an objective type paper of 150 questions, John got 80% correct answers and
Mohan got 64% correct answers.

(i) How many correct answers did each get ?

(ii) What percent is Mohan's correct answers to John's correct answers ?

Answer

Given:

Total questions = 150

Percentage of correct answers for John = 80%

Percentage of correct answers for Mohan = 64%

(i) Number of correct answers John got = 80% of the total questions

$$\begin{aligned} &= \frac{80}{100} \times 150 \\ &= \frac{12000}{100} \\ &= 120 \end{aligned}$$

Number of correct answers Mohan got = 64% of the total questions

$$\begin{aligned} &= \frac{64}{100} \times 150 \\ &= \frac{9600}{100} \\ &= 96 \end{aligned}$$

John got 120 correct answers and Mohan got 96 correct answers.

$$(ii) \text{ Percentage} = \frac{\text{Mohan's correct answers}}{\text{John's correct answers}} \times 100 \%$$

$$= \frac{96}{120} \times 100 \%$$

$$= \frac{9600}{120} \%$$

$$= 80 \%$$

Mohan's correct answers to John's correct are 80 %.

Question 18

The number 8,000 is first increased by 20% and then decreased by 20%. Find the resulting number.

Answer

The original number is 8,000.

After an increase of 20 %, it becomes = 8,000 + 20 % of 8,000

$$= 8,000 + \frac{20}{100} \times 8,000$$

$$= 8,000 + \frac{1,60,000}{100}$$

$$= 8,000 + 1,600$$

$$= 9,600$$

Now it is decreased by 20%, it becomes = 9,600 - 20 % of 9,600

$$= 9,600 - \frac{20}{100} \times 9,600$$

$$= 9,600 - \frac{1,92,000}{100}$$

$$= 9,600 - 1,920$$

$$= 7,680$$

The resulting number = 7,680.

Question 19

The number 12,000 is first decreased by 25% and then increased by 25%. Find the resulting number.

Answer

The original number is 12,000.

After a decrease of 25 %, it becomes = 12,000 - 25 % of 12,000

$$= 12,000 - \frac{25}{100} \times 12,000$$

$$= 12,000 - \frac{3,00,000}{100}$$

$$= 12,000 - 3,000$$

$$= 9,000$$

Now it is increased by 25%, it becomes = 9,000 + 25 % of 9,000

$$= 9,000 + \frac{25}{100} \times 9,000$$

$$= 9,000 + \frac{2,25,000}{100}$$

$$= 9,000 + 2,250$$

$$= 11,250$$

The resulting number = 11,250.

Question 20

The cost of an article is first increased by 20% and then decreased by 30%, find the percentage change in the cost of the article.

Answer

Let the cost of the article be 100.

After an increase of 20 %, it becomes = $100 + 20 \% \text{ of } 100$

$$= 100 + \frac{20}{100} \times 100$$

$$= 100 + \frac{2000}{100}$$

$$= 100 + 20$$

$$= 120$$

Now it is decreased by 30%, it becomes = $120 - 30 \% \text{ of } 120$

$$= 120 - \frac{30}{100} \times 120$$

$$= 120 - \frac{3,600}{100}$$

$$= 120 - 36$$

$$= 84$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 84 - 100$$

$$= -16$$

Change on the whole is negative means change on the whole is decreasing.

$$\therefore \text{Percentage change} = \frac{\text{Change on the whole}}{\text{Initial value}} \times 100 \%$$

$$= \frac{16}{100} \times 100 \%$$

$$= \frac{1600}{100} \%$$

$$= 16\%$$

Hence, the cost of the article is decreased by 16%.

Question 21

The cost of an article is first decreased by 25% and then further decreased by 40%. Find the percentage change in the cost of the article.

Answer

Let the cost of the article be 100.

After a decrease of 25%, it becomes = $100 - 25\% \text{ of } 100$

$$= 100 - \frac{25}{100} \times 100$$

$$= 100 - \frac{2500}{100}$$

$$= 100 - 25$$

$$= 75$$

Now it is decreased by 40%, it becomes = 75 - 40 % of 75

$$= 75 - \frac{40}{100} \times 75$$

$$= 75 - \frac{3,000}{100}$$

$$= 75 - 30$$

$$= 45$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 45 - 100$$

$$= -55$$

Change on the whole is negative means change on the whole is decreasing.

$$\therefore \text{Percentage change} = \frac{\text{Change on the whole}}{\text{Initial value}} \times 100 \%$$

$$= \frac{55}{100} \times 100 \%$$

$$= \frac{5500}{100} \%$$

$$= 55\%$$

Hence, the cost of the article is decreased by 55%.

Exercise 7(B)

Question 1(i)

Out of two students A and B, A does 10 questions and B does 30 questions in the same time. The percentage of number of questions done by B to the number of questions done by A is :

1. 30%
2. 300%
3. 25%
4. $33\frac{1}{3}$

Answer**Given:**

Number of question done by A = 10

Number of question done by B = 30

$$\text{Percentage} = \frac{\text{No. of ques done by B}}{\text{No. of ques done by A}} \times 100\%$$

$$= \frac{30}{10} \times 100\%$$

$$= \frac{3000}{10} \%$$

$$= 300\%$$

Hence, option 2 is the correct option.

Question 1(ii)

In an election, there are only two candidates A and B. A gets 60% of the total votes polled and wins the election by 960 votes. What is the number of total votes polled ?

1. 40%
2. 6400
3. 3200
4. 4800

Answer

Given:

Votes polled by A = 60% of the total votes.

A wins the election by = 960 votes.

Votes polled by B = (100% - 60%) of the total votes = 40% of the total votes.

Let x be the total number of votes. Hence,

Votes polled by A - Votes polled by B = 960

$\Rightarrow$ 60% of the total votes - 40% of the total votes = 960

$\Rightarrow$ 60% of x - 40% of x = 960

$\Rightarrow$ (60% - 40%) of x = 960

$\Rightarrow$ (20%) of x = 960

$\Rightarrow \frac{20}{100} \times x = 960$

$\Rightarrow \frac{1}{5} \times x = 960$

$\Rightarrow x = \frac{960 \times 5}{1}$

$\Rightarrow x = 4,800$

Hence, option 4 is the correct option.

Question 1(iii)

If A is 20% less than B, then B is :

1. 20% more than A
2. 25% less than A
3. 20% less than A
4. 25% more than A

Answer

Let the value of B be 100.

A is 20% less than B.

$$\therefore A = 100 - 20\% \times 100$$

$$\Rightarrow A = 100 - \frac{20}{100} \times 100$$

$$\Rightarrow A = 100 - 20$$

$$\Rightarrow A = 80$$

$$\text{B's percentage} = \frac{\text{B's value} - \text{A's value}}{\text{A's value}} \times 100\%$$

$$= \frac{(100 - 80)}{80} \times 100\%$$

$$= \frac{20}{80} \times 100\%$$

$$= \frac{1}{4} \times 100\%$$

$$= \frac{100}{4}\%$$

$$= 25\%$$

Hence, option 4 is the correct option.

Question 1(iv)

A student has to obtain 35% of the total marks to pass. He got 25% of the total marks and failed by 80 marks. The total of marks is :

1. 400

2. 800

3. 600

4. 750

Answer**Given:**

Passing marks = 35% of the total marks

A candidate gets = 25% of the total marks

The candidate fails by = 80 marks

Let x be the maximum marks.

$$\therefore \text{Passing marks} = \frac{35}{100} \times x$$

$$= \frac{7}{20} \times x$$

$$= \frac{7x}{20} \dots\dots\dots(1)$$

$$\text{A candidate gets} = \frac{25}{100} \times x$$

$$= \frac{1}{4} \times x$$

$$= \frac{x}{4}$$

Since the candidate got $\frac{x}{4}$ marks and fails by 80 marks. Hence, we can say that
passing marks = $\left(\frac{x}{4} + 80\right)$ marks.

Using equation (1), we get,

$$\therefore \frac{7x}{20} = \frac{x}{4} + 80$$

$$\Rightarrow \frac{7x}{20} - \frac{x}{4} = 80$$

LCM of 20 and 4 is 20,

$$\therefore \frac{7x}{20} - \frac{5x}{5 \times 4} = 80$$

$$\Rightarrow \frac{7x}{20} - \frac{5x}{20} = 80$$

$$\Rightarrow \frac{(7-5)}{20} \times x = 80$$

$$\Rightarrow \frac{2}{20} \times x = 80$$

$$\Rightarrow x = \frac{80 \times 20}{2}$$

$$\Rightarrow x = \frac{1600}{2}$$

$$\Rightarrow x = 800$$

Hence, option 2 is the correct option.

Question 1(v)

A mixture of milk and water contains 4 parts of milk and 1 part of water. The percentage of milk in the mixture is :

1. 25%
2. 50%
3. 20%
4. 80%

Answer

Given:

Mixture contains milk = 4 parts

Mixture contains water = 1 parts

Total mixture = 4 + 1 = 5 parts

$$\text{Percentages of milk} = \frac{\text{Mixture contains milk}}{\text{Total mixture}} \times 100\%$$

$$= \frac{4}{5} \times 100\%$$

$$= \frac{400}{5} \%$$

$$= 80\%$$

Hence, option 4 is the correct option.

Question 2

A man bought a certain number of oranges; out of which 13 percent were found rotten. He gave 75% of the remaining in charity and still had 522 oranges left. Find how many oranges had he bought ?

Answer

Let the number of oranges the man bought be x .

Percentage of rotten oranges = 13%

Percentage of remaining oranges = $(100\% - 13\%) = 87\%$

No. of remaining oranges = 87% of x

$$\begin{aligned} &= \frac{87}{100} \times x \\ &= \frac{87x}{100} \dots\dots\dots(1) \end{aligned}$$

Oranges given to charity = 75% of the remaining oranges.

Left over oranges = $(100\% - 75\%) = 25\%$ of the remaining oranges = 522

From equation (1), we get

$$\begin{aligned} \Rightarrow 25\% \text{ of } \frac{87x}{100} &= 522 \\ \Rightarrow \frac{25}{100} \times \frac{87x}{100} &= 522 \\ \Rightarrow \frac{1}{4} \times \frac{87x}{100} &= 522 \\ \Rightarrow \frac{1 \times 87x}{4 \times 100} &= 522 \end{aligned}$$

$$\Rightarrow \frac{87x}{400} = 522$$

$$\Rightarrow x = \frac{522 \times 400}{87}$$

$$\Rightarrow x = \frac{2,08,800}{87}$$

$$\Rightarrow x = 2,400$$

Hence, the total number of oranges = 2,400.

Question 3

5% pupil in a town died due to some disease and 3% of the remaining left the town. If 2,76,450 pupil are still in the town, find the original number of pupil in the town.

Answer

Let the original number of pupil in the town be x .

Percentage of pupil died in the town = 5%

Percentage of remaining pupil in the town = $(100\% - 5\%) = 95\%$

$$\text{Number of remaining pupil in the town} = \frac{95}{100} \times x$$

$$= \frac{19}{20} \times x$$

$$= \frac{19x}{20}$$

Percentage of pupil who left the town = 3% of the remaining pupil

Percentage of pupil still in the town = $(100\% - 3\%) = 97\%$ of the remaining pupil

Number of pupil still in the town = 97% of $\frac{19x}{20}$

Given, number of pupil still in the town = 2,76,450

$$\therefore 97\% \text{ of } \frac{19x}{20} = 2,76,450$$

$$\Rightarrow \frac{97}{100} \times \frac{19x}{20} = 2,76,450$$

$$\Rightarrow \frac{97 \times 19x}{100 \times 20} = 2,76,450$$

$$\Rightarrow \frac{1843x}{2000} = 2,76,450$$

$$\Rightarrow x = \frac{2,76,450 \times 2000}{1843}$$

$$\Rightarrow x = \frac{55,29,00,000}{1843}$$

$$\Rightarrow x = 3,00,000$$

The original number of pupil in the town = 3,00,000.

Question 4

In a combined test in English and Physics; 36% candidates failed in English; 28% failed in Physics and 12% in both; find :

(i) the percentage of passed candidates.

(ii) the total number of candidates appeared, if 208 candidates have failed.

Answer

(i) Percentage of candidates failed only in English = $36\% - 12\% = 24\%$

Percentage of candidates failed only in Physics = $28\% - 12\% = 16\%$

Percentage of candidates failed in both subjects = 12%

Total failed candidates = $24\% + 16\% + 12\% = 52\%$

Total passed candidates = $(100\% - 52\%) = 48\%$

The percentage of passed candidates = 48%.

(ii) Let the total number of candidates be x .

Percentage of failed candidates = 52%

Given, 208 candidates failed,

$\therefore 52\% \text{ of } x = 208$

$$\Rightarrow \frac{52}{100} \times x = 208$$

$$\Rightarrow \frac{13}{25} \times x = 208$$

$$\Rightarrow x = \frac{208 \times 25}{13}$$

$$\Rightarrow x = \frac{5200}{13}$$

$$\Rightarrow x = 400$$

Hence, total number of candidates = 400.

Question 5

In a combined test in Maths and Chemistry, 84% candidates passed in Maths, 76% in Chemistry and 8% failed in both. Find :

(i) the percentage of failed candidates.

(ii) if 340 candidates passed in the test, then, how many candidates had appeared in the test ?

Answer

(i) Percentage of candidates passed in Mathematics = 84%

Percentage of candidates failed in Mathematics = $(100\% - 84\%) = 16\%$

Percentage of candidates failed only in Mathematics = $16\% - 8\% = 8\%$

Percentage of candidates passed in Chemistry = 76%

Percentage of candidates failed in Chemistry = $(100\% - 76\%) = 24\%$

Percentage of candidates failed only in Chemistry = $24\% - 8\% = 16\%$

Percentage of candidates failed in both subjects = 8%

Percentage of failed candidates = $8\% + 16\% + 8\% = 32\%$

The percentage of failed candidates = 32%.

(ii) Percentage of passed candidates = $(100\% - 32\%) = 68\%$

Given, 340 candidates passed,

$\therefore 68\% \text{ of } x = 340$

$$\Rightarrow \frac{68}{100} \times x = 340$$

$$\Rightarrow \frac{17}{25} \times x = 340$$

$$\Rightarrow x = \frac{340 \times 25}{17}$$

$$\Rightarrow x = \frac{8500}{17}$$

$$\Rightarrow x = 500$$

Hence, total number of candidates appeared = 500.

Question 6

A's income is 25% more than B's. Find out by how much percent is B's income less than A's ?

Answer

Lets take B's income to be 100.

A's income is 25% more than B's income.

Hence,

$$A = 100 + 25$$

$$\Rightarrow A = 100 + \frac{25}{100} \times 100$$

$$\Rightarrow A = 100 + 25$$

$$\Rightarrow A = 125$$

If A's income is 125, B's income = 25 less than A's.

$$\begin{aligned}\text{B's percentage} &= \frac{\text{A's income} - \text{B's income}}{\text{A's income}} \times 100\% \\&= \frac{(125 - 100)}{125} \times 100\% \\&= \frac{25}{125} \times 100\% \\&= \frac{1}{5} \times 100\% \\&= \frac{100}{5}\% \\&= 20\%\end{aligned}$$

Hence, B's income is 20% less than A's.

Question 7

Mona is 20% younger than Neetu. By how much percent is Neetu older than Mona ?

Answer

Lets take Neetu's age to be 100 years.

Mona is 20% younger than Neetu.

$\therefore$ Mona's age = 100 - 20% of 100

$$\begin{aligned}&= 100 - \frac{20}{100} \times 100 \\&= 100 - 20 \\&= 80\end{aligned}$$

Mona's age is 80 years and Neetu's age is 100 years. Then we can say that Neetu's age is 20 years more than Mona's age.

Percentage by which Neetu is older than Mona =

$$\frac{\text{Neetu's age} - \text{Mona's age}}{\text{Mona's age}} \times 100\%$$

$$= \frac{(100 - 80)}{80} \times 100\%$$

$$= \frac{20}{80} \times 100\%$$

$$= \frac{1}{4} \times 100\%$$

$$= \frac{100}{4}\%$$

$$= 25\%$$

Hence, Neetu is older than Mona by 25%.

Question 8

If the price of sugar is increased by 25% today, by what percent should it be decreased tomorrow to bring the price back to the original ?

Answer

Lets take today's price of sugar as ₹100.

The price of sugar is increased by 25%.

$$\therefore \text{Increased Price} = 100 + 25\% \text{ of } 100$$

$$= 100 + \left(\frac{25}{100} \times 100 \right)$$

$$= 100 + \left(\frac{1}{4} \times 100 \right)$$

$$= 100 + \left(\frac{100}{4} \right)$$

$$= 100 + 25$$

$$= ₹125$$

If new price is ₹125 and old price was ₹100, then we can say that new price is ₹25 more than the old price.

$$\text{Percentage decrease} = \frac{\text{New price} - \text{old price}}{\text{new price}} \times 100\%$$

$$= \frac{(125 - 100)}{125} \times 100\%$$

$$= \frac{25}{125} \times 100\%$$

$$= \frac{1}{5} \times 100\%$$

$$= \frac{100}{5}\%$$

$$= 20\%$$

Hence, price should be decreased 20% tomorrow to bring the price back to the original.

Question 9

A number increased by 15% becomes 391. Find the number.

Answer

Let the number be x .

The number increased by 15%.

Hence,

$$\Rightarrow x + 15\% \text{ of } x = 391$$

$$\Rightarrow x + \frac{15}{100} \times x = 391$$

$$\Rightarrow x + \frac{3}{20} \times x = 391$$

$$\Rightarrow \frac{20}{20} \times x + \frac{3}{20} \times x = 391$$

$$\Rightarrow \frac{(20 + 3)}{20} \times x = 391$$

$$\Rightarrow \frac{23}{20} \times x = 391$$

$$\Rightarrow x = \frac{391 \times 20}{23}$$

$$\Rightarrow x = \frac{7,820}{23}$$

$$\Rightarrow x = 340$$

Hence, the number is 340.

Question 10

A number decreased by 23% becomes 539. Find the number.

Answer

Let the number be x .

The number decreased by 23%.

Hence,

$$\Rightarrow x - 23\% \text{ of } x = 539$$

$$\Rightarrow x - \frac{23}{100} \times x = 539$$

$$\Rightarrow \frac{100}{100} \times x - \frac{23}{100} \times x = 539$$

$$\Rightarrow \frac{(100 - 23)}{100} \times x = 539$$

$$\Rightarrow \frac{77}{100} \times x = 539$$

$$\Rightarrow x = \frac{539 \times 100}{77}$$

$$\Rightarrow x = \frac{53,900}{77}$$

$$\Rightarrow x = 700$$

Hence, the number is 700.

Question 11

Two numbers are respectively 20 percent and 50 percent more than a third number. What percent of the first number is the second number?

Answer**Given:**

First number is 20% more than the third number.

Second number is 50% more than the third number.

Lets take the third number to be 100.

First number = $100 + 20\%$ of 100

$$= 100 + \left(\frac{20}{100} \times 100 \right)$$

$$= 100 + \left(\frac{20}{100} \times 100 \right)$$

$$= 100 + 20$$

$$= 120$$

Second number = $100 + 50\%$ of 100

$$= 100 + \left(\frac{50}{100} \times 100 \right)$$

$$= 100 + \left(\frac{50}{100} \times 100 \right)$$

$$= 100 + 50$$

$$= 150$$

$$\text{Percentage of first number is to second number} = \frac{\text{Second number}}{\text{First number}} \times 100 \%$$

$$= \frac{150}{120} \times 100 \%$$

$$= \frac{5}{4} \times 100 \%$$

$$= \frac{500}{4} \%$$

$$= 125\%$$

Hence, first number is 125% of the second number.

Question 12

Two numbers are respectively 20 percent and 50 percent of a third number.

What percent of the first number is the second number ?

Answer

Given:

First number is 20% of the third number.

Second number is 50% of the third number.

Lets take the third number to be 100.

First number = 20% of 100

$$= \frac{20}{100} \times 100$$

$$= \frac{20}{100} \times 100$$

$$= 20$$

Second number = 50% of 100

$$= \frac{50}{100} \times 100$$

$$= \frac{50}{\cancel{100}} \times \cancel{100}$$

$$= 50$$

$$\text{Percentage of first number is to second number} = \frac{\text{Second number}}{\text{First number}} \times 100 \%$$

$$= \frac{50}{20} \times 100 \%$$

$$= \frac{5}{2} \times 100 \%$$

$$= \frac{500}{2} \%$$

$$= 250\%$$

Hence, the percent of the first number is the second number = 250 %.

Question 13

Two numbers are respectively 30 percent and 40 percent less than a third number. What percent of the first number is the second number ?

Answer

Given:

First number is 30% less than the third number.

Second number is 40% less than the third number.

Lets take the third number to be 100.

First number = $100 - 30\%$ of 100

$$= 100 - \frac{30}{100} \times 100$$

$$= 100 - \frac{30}{\cancel{100}} \times \cancel{100}$$

$$= 100 - 30$$

$$= 70$$

Second number = $100 - 40\%$ of 100

$$= 100 - \frac{40}{100} \times 100$$

$$= 100 - \frac{40}{\cancel{100}} \times \cancel{100}$$

$$= 100 - 40$$

$$= 60$$

Percentage of first number is to second number = $\frac{\text{Second number}}{\text{First number}} \times 100 \%$

$$= \frac{60}{70} \times 100 \%$$

$$= \frac{6}{7} \times 100 \%$$

$$= \frac{600}{7} \%$$

$$= 85\frac{5}{7} \%$$

Hence, the percent of the first number is the second number = $85\frac{5}{7}\%$.

Question 14

Mohan gets ₹ 1,350 from Geeta and ₹ 650 from Rohit. Out of the total money that Mohan gets from Geeta and Rohit, what percent does he get from Rohit ?

Answer

Given:

Mohan gets ₹ 1,350 from Geeta

Mohan gets ₹ 650 from Rohit

Total money Mohan gets = ₹ (1,350 + 650)

= ₹ 2,000

Percentage of money Mohan got from Rohit =

$$\frac{\text{Money Mohan gets from Rohit}}{\text{Total money Mohan gets}} \times 100\%$$

$$= \frac{650}{2000} \times 100\%$$

$$= \frac{13}{40} \times 100\%$$

$$= \frac{1,300}{40}\%$$

$$= 32.5\%$$

Hence, Mohan gets 32.5% of the total money from Rohit.

Question 15

The monthly income of a man is ₹ 16,000. 15 percent of it is paid as income-tax and 75% of the remainder is spent on rent, food, clothing, etc. How much money is still left with the man ?

Answer**Given:**

The monthly income of the man = ₹ 16,000

Percent of money the man paid as income tax = 15% of the salary

Remaining percent of money after paying income tax = (100% - 15%) of the salary = 85% of the salary

$$= \frac{85}{100} \times 16,000$$

$$= \frac{17}{20} \times 16,000$$

$$= \frac{2,72,000}{20}$$

$$= ₹ 13,600$$

Percent of money the man spent on rent, food, clothing, etc = 75% of the remaining amount

Left over money = (100% - 75%) of the remaining amount = 25% of the remaining amount.

$$= \frac{25}{100} \times 13,600$$

$$= \frac{1}{4} \times 13,600$$

$$= \frac{13,600}{4}$$

$$= ₹ 3,400$$

Money still left with the man = ₹ 3,400

Question 16

During 2003, the production of a factory decreased by 25%. But during 2004, it (production) increased by 40% of what it was at the beginning of 2004.

Calculate the resulting change (increase or decrease) in production during these two years.

Answer

Let the production of a factory at the beginning of 2003 be 100.

The production of a factory decreased by 25%.

Hence,

Decreased production in 2003 = 100 - 25% of 100

$$= 100 - \left(\frac{25}{100} \times 100 \right)$$

$$= 100 - \left(\frac{25}{100} \times 100 \right)$$

$$= 100 - 25$$

$$= 75$$

During 2004, production increased by 40%.

Hence,

Increased production in 2004 = 75 + 40% of 75

$$= 75 + \left(\frac{40}{100} \times 75 \right)$$

$$= 75 + \left(\frac{2}{5} \times 75 \right)$$

$$= 75 + \frac{150}{5}$$

$$= 75 + 30$$

$$= 105$$

Resulting change = Final production - Initial production

$$= 105 - 100$$

$$= 05$$

Resulting change is positive means production is increased.

$$\text{Percentage change} = \frac{\text{Resulting change}}{\text{Initial production}} \times 100 \%$$

$$= \frac{5}{100} \times 100 \%$$

$$= \frac{5}{\cancel{100}} \times \cancel{100} \%$$

$$= 5 \%$$

Hence, the increase in production is 5%.

Question 17

Last year, oranges were available at ₹ 24 per dozen; but this year, they are available at ₹ 50 per score. Find the percentage change in the price of oranges.

[1 score = 20]

Answer

Last year, cost of oranges = ₹ 24 per dozen

$$\text{Cost of one orange} = ₹ \frac{24}{12} = ₹ 2$$

This year, cost of oranges = ₹ 50 per score

$$\text{Cost of one orange} = ₹ \frac{50}{20} = ₹ 2.5$$

$$\text{Change in price} = ₹ (2.5 - 2) = ₹ 0.5$$

Change in price is positive means price of oranges has increased.

$$\text{Percentage increase} = \frac{\text{Change in price}}{\text{Cost of orange last year}} \times 100 \%$$

$$= \frac{0.5}{2} \times 100 \%$$

$$= \frac{50}{2} \%$$

$$= 25 \%$$

Hence, percentage change in the price of orange = 25% increase.

Question 18(i)

Increase 180 by 25%.

Answer

$$\begin{aligned} & 180 + 25\% \text{ of } 180 \\ &= 180 + \frac{25}{100} \times 180 \\ &= 180 + \frac{1}{4} \times 180 \\ &= 180 + \frac{180}{4} \\ &= 180 + 45 \\ &= 225 \end{aligned}$$

Hence, on increasing 180 by 25% we get 225.

Question 18(ii)

Decrease 140 by 18%.

Answer

$$\begin{aligned} & 140 - 18\% \text{ of } 140 \\ &= 140 - \frac{18}{100} \times 140 \\ &= 140 - \frac{9}{50} \times 140 \\ &= 140 - \frac{1,260}{50} \end{aligned}$$

$$= 140 - 25.2$$

$$= 114.8$$

Hence, on decreasing 140 by 18% we get 114.8.

Question 19(i)

A number when increased by 23% becomes 861; find the number.

Answer

Let the number be x .

The number is increased by 23%.

$\therefore$ Amount by which the number is increased = 23% of x

$$= \frac{23}{100} \times x$$

$$= \frac{23x}{100}$$

Given, the increased number becomes 861,

$$\therefore x + \frac{23x}{100} = 861$$

$$\Rightarrow \frac{100x + 23x}{100} = 861$$

$$\Rightarrow \frac{123x}{100} = 861$$

$$\Rightarrow x = \frac{861 \times 100}{123}$$

$$\Rightarrow x = \frac{86,100}{123}$$

$$\Rightarrow x = 700$$

Hence, the number is 700.

Question 19(ii)

A number when decreased by 16% becomes 798; find the number.

Answer

Let the number be x .

The number is decreased by 16%.

$\therefore$ Amount by which the number is decreased = 16% of x

$$= \frac{16}{100} \times x$$

$$= \frac{16x}{100}$$

Given, the decreased number becomes 798,

$$\therefore x - \frac{16x}{100} = 798$$

$$\Rightarrow \frac{100x - 16x}{100} = 798$$

$$\Rightarrow \frac{84x}{100} = 798$$

$$\Rightarrow x = \frac{798 \times 100}{84}$$

$$\Rightarrow x = \frac{79,800}{84}$$

$$\Rightarrow x = 950$$

Hence, the number is 950.

Question 20

The price of sugar is increased by 20%. By what percent must the consumption of sugar be decreased so that the expenditure on sugar may remain the same ?

Answer

Lets take the price of sugar to be ₹ 100 and consumption be 1 kg.

Hence, total expenditure will be

$$100 \times 1 = 100 \dots\dots\dots(1)$$

The price of sugar is increased by 20%. Hence,

$$\text{New Price} = 100 + 20\% \times 100$$

$$= 100 + \frac{20}{100} \times 100$$

$$= 100 + \frac{20}{100} \times \cancel{100}$$

$$= 100 + 20$$

$$= ₹ 120$$

Let's take new consumption to be x kg.

$$\text{New expenditure} = ₹ (120 \times x)$$

Given that expenditure should remain same,

$$\therefore 120x = 100$$

$$\Rightarrow x = \frac{100}{120}$$

$$\Rightarrow x = \frac{5}{6}$$

$$\text{Percentage decrease in consumption} = \frac{\text{Old Value} - \text{New Value}}{\text{Old Value}} \times 100$$

$$= \frac{1 - \frac{5}{6}}{1} \times 100 = \frac{6 - 5}{6} \times 100 = \frac{1}{6} \times 100 = \frac{100}{6} = \frac{50}{3} = 16\frac{2}{3}$$

Hence, consumption should be decreased by $16\frac{2}{3}\%$.

Test Yourself

Question 1(i)

A number, whose 4% is 6, is :

1. 24

2. 0.24

3. 150

4. 75

Answer

Let the number be x .

$$4 \Rightarrow \frac{4}{100} \times x = 6$$

$$\Rightarrow \frac{1}{25} \times x = 6$$

$$\Rightarrow x = \frac{6 \times 25}{1}$$

$$\Rightarrow x = 150$$

Hence, option 3 is the correct option.

Question 1(ii)

What percent of 50 is 10 ?

1. 20%

2. 500%

3. 500

4. 20

Answer

Let the percentage be x

$$x \Rightarrow \frac{x}{100} \times 50 = 10$$

$$\Rightarrow \frac{50x}{100} = 10$$

$$\Rightarrow \frac{x}{2} = 10$$

$$\Rightarrow x = \frac{10 \times 2}{1}$$

$$\Rightarrow x = 20$$

Hence, option 1 is the correct option.

Question 1(iii)

18 hours as a percentage of 3 days is :

1. $\frac{18}{3} \times 100$

2. $\frac{3}{18} \times 100$

3. $\frac{18}{3 \times 24} \times 100$

4. $\frac{3 \times 24}{18} \times 100$

Answer

Number of days = 3

Hours in 1 day = 24 hours

Total hours in 3 days = 3 x 24 hours

$$\begin{aligned} \text{18 hours as a percentage of 3 days} &= \frac{\text{Given hours}}{\text{Total hours}} \times 100\% \\ &= \frac{18}{3 \times 24} \times 100\% \end{aligned}$$

Hence, option 3 is the correct option.

Question 1(iv)

An alloy contains 30% of copper, 30% of zinc and rest nickel. The amount of nickel in 400 gm of alloy is :

1. 40% of 400 gm
2. 30% of 400 gm
3. 70% of 400 gm
4. 400 gm - 30% of 400 gm

Answer

Total weight of alloy = 400gm

Percentage of copper in alloy = 30% of 400gm

Percentage of zinc in alloy = 30% of 400gm

Percentage of copper and zinc in alloy = (30% + 30%) of 400gm

= 60% of 400gm

Percentage of nickel in alloy = (100% - 60%) of 400 gm

= 40% of 400gm

Hence, option 1 is the correct option.

Question 1(v)

A number is first decreased by 40% and then increased by 40%. The equivalent change is :

1. nothing
2. 16% increase
3. 16% decrease
4. 8% increase

Answer

Let the original number be 100.

After a decrease of 40%, it becomes = 100 - 40% of 100

$$= 100 - \frac{40}{100} \times 100$$

$$= 100 - \frac{4000}{100}$$

$$= 100 - 40$$

$$= 60$$

Now it is increased by 40%, it becomes = 60 + 40% of 60

$$= 60 + \frac{40}{100} \times 60$$

$$= 60 + \frac{2,400}{100}$$

$$= 60 + 24$$

$$= 84$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 84 - 100$$

$$= -16$$

Change on the whole is negative means change on the whole the number is decreasing.

$$\therefore \text{Percentage change} = \frac{\text{Change on the whole}}{\text{Initial value}} \times 100 \%$$

$$= \frac{16}{100} \times 100 \%$$

$$= 16\%$$

Hence, option 3 is the correct option.

Question 1(vi)

Out of 700 eggs, 20% are rotten. The number of good eggs is :

1. 140

2. 560

3. 680

4. 840

Answer

Total number of eggs = 700

Rotten eggs = 20% of eggs

= 20% of 700

$$= \frac{20}{100} \times 700$$

$$= \frac{1}{5} \times 700$$

$$= \frac{700}{5}$$

$$= 140$$

Good eggs = Total eggs - Rotten eggs

$$= 700 - 140$$

$$= 560$$

Hence, option 2 is the correct option.

Question 1(vii)

80% of 200 - 50 is equal to :

1. 30% of 200
2. 110
3. 80% of 150
4. none of these

Answer

80% of 200 - 50

$$= \frac{80}{100} \times 200 - 50$$

$$= \frac{80}{100} \times 200 - 50$$

$$= (80 \times 2) - 50$$

$$= 160 - 50$$

$$= 110$$

Hence, option 2 is the correct option.

Question 1(viii)

A number 80 is wrongly taken as 100. The percentage error is :

1. 20%
2. 25%

$$3. \frac{100}{80} \times 100$$

$$4. \frac{80}{100} \times 100$$

Answer

Given:

Correct Number = 80

Incorrect Number = 100

Difference in number = Incorrect number - Correct number

$$= 100 - 80 = 20$$

$$\text{Percentage error} = \frac{\text{Difference in number}}{\text{Correct Number}} \times 100 \%$$

$$= \frac{20}{80} \times 100 \%$$

$$= \frac{1}{4} \times 100 \%$$

$$= 25\%$$

Hence, option 2 is the correct option.

Question 1(ix)

The price of an article was ₹ 680 last year. This year its price is ₹ 816. The percentage change in the price is :

1. $(816 - 680)\%$ increases

2. $(816 - 680)\%$ decreases

$$3. \frac{816 - 680}{680} \times 100\% \text{ increases}$$

$$4. \frac{816 - 680}{816} \% \text{ increases}$$

Answer

Given:

Price of an article last year = ₹ 680

Price of an article this year = ₹ 816

Difference in price = Price of an article this year - Price of an article last year

= ₹ 816 - ₹ 680

$$\text{Percentage change in the price} = \frac{\text{Difference in price}}{\text{Price of an article last year}} \times 100 \%$$

$$= \frac{(816 - 680)}{680} \times 100 \%$$

Price of the article this year is greater than price of the article last year, means percentage change will increase.

Hence, option 3 is the correct option.

Question 1(x)

Statement 1: To change a number in percentage to a ratio, write it as fraction with denominator 100 and then reduce it to the lowest term if possible.

Statement 2: A percentage can be converted to a fraction by removing sign of % dividing by 100.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

According to statement 1 :

To change a number in percentage to a ratio, write it as fraction with denominator 100 and then reduce it to the lowest term if possible.

Lets take an example;

$$60\% = \frac{60}{100} = \frac{6}{10} = \frac{3}{5} = 3 : 5$$

∴ Statement 1 is true.

According to statement 2 :

A percentage can be converted to a fraction by removing sign of % dividing by 100.

Lets take an example;

$$40 = \frac{40}{100} = \frac{2}{5}$$

So, statement 2 is true.

Hence, option 1 is the correct option.

Question 1(xi)

Assertion (A) : 9 % of $\sqrt{0.0169}$ is 0.0117.

Reason (R) : To find a percentage of a quantity, we change the percentage to a fraction or a decimal and multiply it by the quantity.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Solving,

$$\Rightarrow 9 \Rightarrow \frac{9}{100} \times \sqrt{0.0169}$$

$$\Rightarrow \frac{9}{100} \times 0.13$$

$$\Rightarrow \frac{1.17}{100}$$

$$\Rightarrow 0.0117$$

So, assertion (A) is true.

To find a percentage of a quantity we change the percentage to fraction or a decimal and multiply it by the quantity.

For example; 20% of a.

$$= \frac{20}{100} \times a$$

$$= 0.2 \times a$$

$$= 0.2a$$

So, reason (R) is true and reason clearly explains assertion.

Hence, option 1 is the correct option.

Question 1(xii)

Assertion (A) : If we decrease ₹ 120 by $12\frac{1}{2}$, then decreased amount = ₹ 105.

Reason (R) : Percentage change =

$$\left(\frac{\text{Actual change (increase/decrease)}}{\text{Original quantity}} \times 100 \right).$$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

$$\Rightarrow 12\frac{1}{2} \Rightarrow \frac{25}{100} \times 120$$

$$\Rightarrow \frac{25}{200} \times 120$$

$$\Rightarrow \frac{25 \times 3}{5}$$

$$\Rightarrow 5 \times 3$$

$$\Rightarrow 15.$$

Decreased amount = ₹ 120 - ₹ 15 = ₹ 105

So, assertion (A) is true.

$$\text{Percentage change} = \left(\frac{\text{Actual change (increase/decrease)}}{\text{Original quantity}} \times 100 \right)$$

If a quantity changes by 15 out of 120, then:

$$\text{Percentage change} = \left(\frac{15}{120} \times 100 \right) = \frac{150}{12} = 12.5$$

So, reason (R) is true and reason clearly explains assertion.

Hence, option 1 is the correct option.

Question 1(xiii)

Assertion (A) : By increasing ₹ 320 by 20%, we obtain the increased amount = ₹ 348.

Reason (R) : To increase a quantity by a given percentage, we first find the percentage of the quantity and then add it to the original quantity.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

$$20\% \text{ of ₹ } 320 = \frac{20}{100} \times 320 = ₹ 64$$

$$\text{Increased amount} = ₹ 320 + ₹ 64 = ₹ 384$$

So, assertion (A) is false.

We know that,

To increase a quantity by a percentage, we first find the percentage of the quantity and then add it to the original quantity.

So, reason (R) is true.

Hence, option 4 is the correct option.

Question 1(xiv)

Assertion (A) : The sum of two numbers is $\frac{28}{25}$ of the first number. Then the second number is 12% of the first number.

Reason (R) : To express one quantity as a percentage of the other we write the other quantity as fraction of the one and then multiply by 100.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Let the first number be x and second number be y .

According to question,

$$\Rightarrow x + y = \frac{28x}{25}$$

$$\Rightarrow y = \frac{28x}{25} - x$$

$$\Rightarrow y = \frac{28x - 25x}{25}$$

$$\Rightarrow y = \frac{3x}{25}$$

Expressing y as the percentage of the first number :

$$\Rightarrow \frac{y}{x} \times 100 \Rightarrow \frac{\frac{3x}{25}}{x} \times 100 \Rightarrow \frac{3}{25} \times 100 \Rightarrow 3 \times 4 \Rightarrow 12$$

So, assertion (A) is true.

According to reason :

To express one quantity as a percentage of the other we write the other quantity as fraction of the first and then multiply by 100.

Lets take an example if we want to know what percent 30 is of 150,

$$\Rightarrow \frac{30}{150} \times 100$$

$$\Rightarrow \frac{1}{5} \times 100$$

$$\Rightarrow 20$$

So, reason (R) is true and reason clearly explains assertion.

Hence, option 1 is the correct option.

Question 2

A family spends 30% of its income on house rent and 60% of the rest on house hold expenses. If the total savings of the family is ₹ 12,600 per month, find the total monthly income of the family.

Answer

Given:

Percent of money the family spends on house rent = 30% of the salary

Remaining percent of money after paying house rent = (100% - 30%) = 70% of the salary

Let the salary of family be x .

$$\begin{aligned}\text{Amount of money left after paying house rent} &= \frac{70}{100} \times x \\ &= \frac{7}{10} \times x \\ &= \frac{7x}{10} \dots\dots\dots(1)\end{aligned}$$

Percent of money the family spent on household expenses = 60% of the remaining amount

Left over money = (100% - 60%) = 40% of the remaining amount = 12,600.

$$\Rightarrow \frac{40}{100} \times \frac{7x}{10} = 12,600$$

$$\Rightarrow \frac{2}{5} \times \frac{7x}{10} = 12,600$$

$$\Rightarrow \frac{2 \times 7x}{5 \times 10} = 12,600$$

$$\Rightarrow \frac{14x}{50} = 12,600$$

$$\Rightarrow \frac{7x}{25} = 12,600$$

$$\Rightarrow x = \frac{12,600 \times 25}{7}$$

$$\Rightarrow x = \frac{3,15,00}{7}$$

$$\Rightarrow x = 45,000$$

Hence, the monthly income of the family = ₹ 45,000

Question 3

Geeta saves 20% of her monthly salary and saves ₹ 43,500 per month. Find her monthly expenditure.

Answer**Given:**

Geeta saves = 20% of her monthly salary

Geeta's saving amount = ₹ 43,500

Let the salary of Geeta be x . Hence,

20% of $x = 43,500$

$$\Rightarrow \frac{20}{100} \times x = 43,500$$

$$\Rightarrow \frac{1}{5} \times x = 43,500$$

$$\Rightarrow x = \frac{43,500 \times 5}{1}$$

$$\Rightarrow x = 2,17,500$$

Geeta's expenditure = (100% - 20%) of her monthly salary

= 80% of her monthly salary

$$= \frac{80}{100} \times 2,17,500$$

$$= \frac{4}{5} \times 2,17,500$$

$$= \frac{8,70,000}{5}$$

$$= 1,74,000$$

Hence, Geeta's monthly expenditure = ₹ 1,74,000.

Question 4

In an examination, 92% of the candidates passed and 96 failed. Find the number of candidates who appeared for this exam.

Answer

Percent of candidates who passed the exam = 92%

Percent of candidates who failed the exam = (100% - 92%) = 8%

Number of candidates who failed the exam = 96

Let the total number of candidates be x .

$$\therefore 8 \Rightarrow \frac{8}{100} \times x = 96$$

$$\Rightarrow x = \frac{96 \times 100}{8}$$

$$\Rightarrow x = \frac{9600}{8}$$

$\Rightarrow x = 1200$ **Hence, total number of candidates who appeared for exam = 1200.**

Question 5

A number is increased by 30% and then this increased number is decreased by 30%. Find the net change.

Answer

Let the original number be 100.

After an increase of 30%, it becomes = $100 + 30\%$ of 100

$$= 100 + \frac{30}{100} \times 100$$

$$= 100 + \frac{30}{100} \times 100$$

$$= 100 + 30$$

$$= 130$$

Now it is decreased by 30%, it becomes = $130 - 30\%$ of 130

$$= 130 - \frac{30}{100} \times 130$$

$$= 130 - \frac{3,900}{100}$$

$$= 130 - 39$$

$$= 91$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 91 - 100$$

$$= -9$$

Change on the whole is negative means the number is decreasing.

$$\begin{aligned}\therefore \text{Percentage change} &= \frac{\text{Change on the whole}}{\text{Initial value}} \times 100 \\ &= \frac{9}{100} \times 100 \\ &= 9\end{aligned}$$

Hence, the net change is decreasing 9%.

Question 6

A number is decreased by 30% and then this decreased number is increased by 30%. Find the net change as percent.

Answer

Let the original number be 100.

After a decrease of 30%, it becomes = 100 - 30% of 100

$$\begin{aligned}&= 100 - \frac{30}{100} \times 100 \\ &= 100 - \frac{3000}{100} \\ &= 100 - 30 \\ &= 70\end{aligned}$$

Now it is increased by 30%, it becomes = 70 + 30% of 70

$$= 70 + \frac{30}{100} \times 70$$

$$= 70 + \frac{2,100}{100}$$

$$= 70 + 21$$

$$= 91$$

$\therefore$ Change on the whole = Final value - Initial value

$$= 91 - 100$$

$$= -9$$

Change on the whole is negative means the number is decreasing.

$$\therefore \text{Percentage change} = \frac{\text{Change on the whole}}{\text{Initial value}} \times 100$$

$$= \frac{9}{100} \times 100$$

$$= 9$$

Hence, the net change is decreasing 9%.

Question 7

The population of a village increases by 10% per year. If the present population of the village is 24,000; find it at the end of 2 years.

Answer

Given:

Percentage increase in village population per year = 10%

Present population of village = 24,000

Increase of population at the end of 1 year = 10% of previous population

$$\begin{aligned} &= \frac{10}{100} \times 24,000 \\ &= \frac{1}{10} \times 24,000 \\ &= \frac{24,000}{10} \\ &= 2,400 \end{aligned}$$

Population at the end of 1 year = Population + Increase of population

$$\begin{aligned} &= 24,000 + 2,400 \\ &= 26,400 \end{aligned}$$

Increase in population at the end of 2 year = 10% of previous population

$$\begin{aligned} &= \frac{10}{100} \times 26,400 \\ &= \frac{1}{10} \times 26,400 \\ &= \frac{26,400}{10} \\ &= 2,640 \end{aligned}$$

Population at the end of 2 year = Previous population + Increase in population

$$\begin{aligned} &= 26,400 + 2,640 \\ &= 29,040 \end{aligned}$$

Hence, the population after 2 year = 29,040

Question 8

The cost of a machine decreases by 10% per year. If its present cost is ₹ 24,000: find its value at the beginning of 3rd year.

Answer**Given:**

The cost of a machine decreases by = 10%

The present cost of a machine = 24,000

Decrease of value at the beginning of 2nd year = 10% of previous value

$$= \frac{10}{100} \times 24,000$$

$$= \frac{1}{10} \times 24,000$$

$$= \frac{24,000}{10}$$

$$= 2,400$$

Value at the beginning of 2nd year = Previous value - Decrease in value

$$= 24,000 - 2,400$$

$$= 21,600$$

Decrease of value at the beginning of 3rd year = 10% of previous value

$$= \frac{10}{100} \times 21,600$$

$$\begin{aligned}
 &= \frac{1}{10} \times 21,600 \\
 &= \frac{21,600}{10} \\
 &= 2,160
 \end{aligned}$$

Value at the beginning of 3rd year = Previous value - Decrease in value

$$\begin{aligned}
 &= 21,600 - 2,160 \\
 &= 19,440
 \end{aligned}$$

Hence, the value at the beginning of 3rd year = 19,440

Question 9

The price of sugar has been increased by 50%. By how much percent can the consumption of the sugar be decreased in order to keep the expenditure on sugar the same.

Answer

Lets take the price of sugar to be ₹ 100 and consumption be 1 kg.

Hence, total expenditure will be

$$100 \times 1 = 100 \dots\dots\dots(1)$$

The price of sugar is increased by 50%. Hence,

$$\text{New Price} = 100 + 50\% \times 100$$

$$= 100 + \frac{50}{100} \times 100$$

$$= 100 + \frac{50}{100} \times 100$$

$$= 100 + 50$$

$$= ₹ 150$$

Let's take new consumption to be x kg.

New expenditure = ₹ $(150 \times x)$

Given that expenditure should remain same,

$$\therefore 150x = 100$$

$$\Rightarrow x = \frac{100}{150}$$

$$\Rightarrow x = \frac{2}{3}$$

$$\text{Percentage decrease in consumption} = \frac{\text{Old Value} - \text{New Value}}{\text{Old Value}} \times 100$$

$$= \frac{1 - \frac{2}{3}}{1} \times 100 = \frac{3 - 2}{3} \times 100 = \frac{1}{3} \times 100 = \frac{100}{3} = 33\frac{1}{3}$$

Hence, consumption should be decreased by $33\frac{1}{3}\%$.

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